Adapting Foundation Models: From Reinforcement Learning to Multivariate Time Series Forecasting

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About Me

- **2016–2020** — MEng, *École des Mines*
  - Major: Data Science

- **2020–2021** — MSc, *ENS Paris-Saclay*
  - Applied Mathematics (MVA Program)
  - Research internship: *InstaDeep* (Model-based RL)

- **2021–2023** — Research Engineer
  - Huawei Noah's Ark Lab (Model-based RL)

- **2023–2026** — PhD, *EURECOM (Sorbonne University) & Huawei Noah's Ark Lab*
  - Thesis topic: Model-based Reinforcement Learning in the era of Foundation Models

Outline

1 Adapting LLMs for Model-based Reinforcement Learning
- Preliminaries
- Problem setup
- Approach
- Results

2 Adapting Time Series Foundation models
- Problem setup
- Approach
- Results

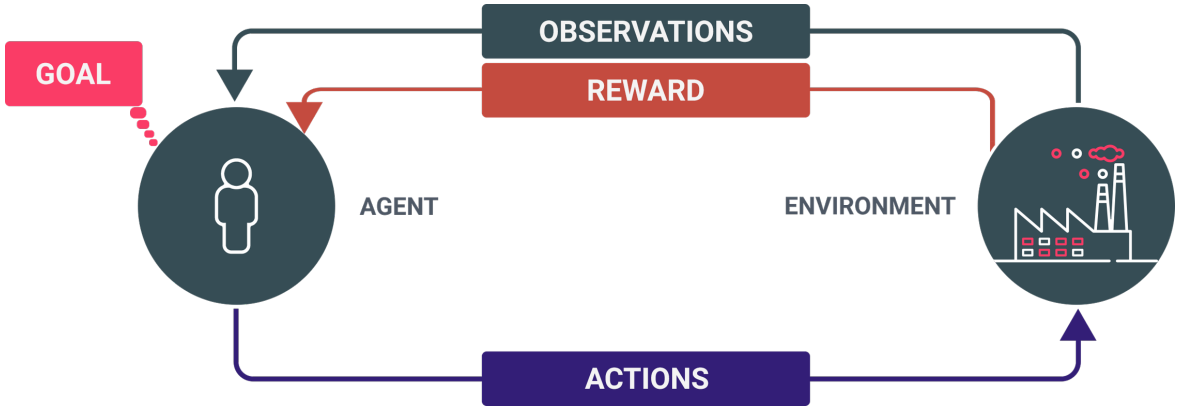
3 Conclusion

Preliminaries

Reinforcement Learning

Reinforcement Learning environments are Markov decision processes $\mathcal{M} = \langle \mathcal{S}, \mathcal{A}, P, r, \mu_0, \gamma \rangle$, where:

- S state space, A action space.
- Transition fn $P_t : (s, a, s') \mapsto \mathbf{Pr}(s_{t+1} = s' | s_t = s, a_t = a)$.
- Reward function $r : (s, a) \mapsto r(s, a)$.
- μ_0 initial state distribution, $\gamma \in [0, 1]$ discount factor.



Reinforcement Learning

The goal of RL is to find a policy $\pi : \mathcal{S} \to \Delta(\mathcal{A})$ that maximizes the return:

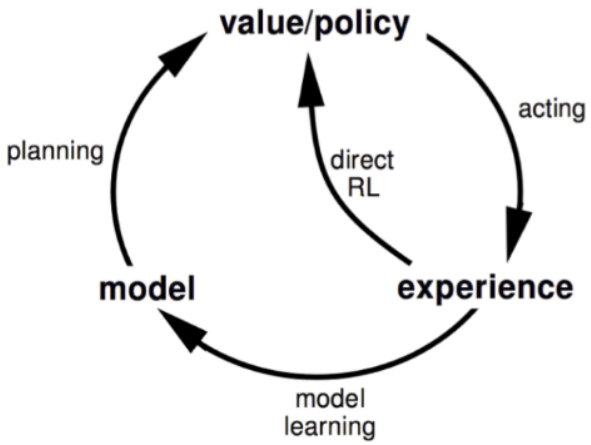
$$\eta(\pi) := \mathbb{E}_{s_0 \sim \mu_0, a_t \sim \pi,\, s_{t>0} \sim P_t}\left[\sum_{t=0}^{\infty} \gamma^t r(s_t, a_t)\right]$$

Model-based Reinforcement Learning

Model-based RL (MBRL) learns the transition $\hat{P}$ from interaction data. The model maximizes the log-likelihood:

$$\mathcal{L}(\mathcal{D}; \hat{P}) = \frac{1}{N} \sum_{i=1}^{N} \log \hat{P}(s_{t+1}^i | s_t^i, a_t^i)$$

The learned model is used for policy search under the *learned* MDP $\widehat{\mathcal{M}} = \langle \mathcal{S}, \mathcal{A}, \hat{P}, r, \mu_0, \gamma \rangle$.



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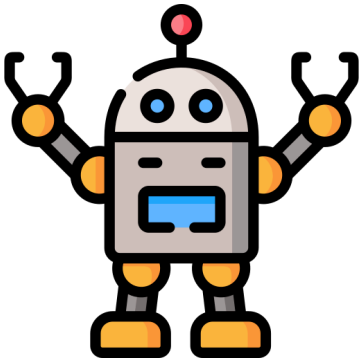
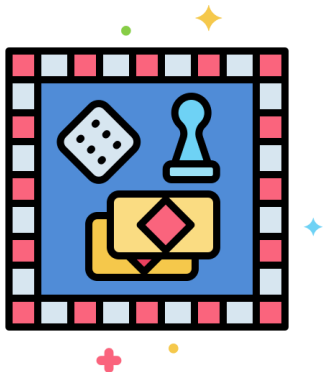
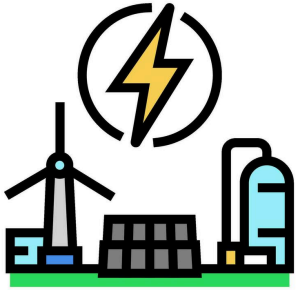
Motivation

MBRL is particularly useful under **budget** and **safety** constraints.

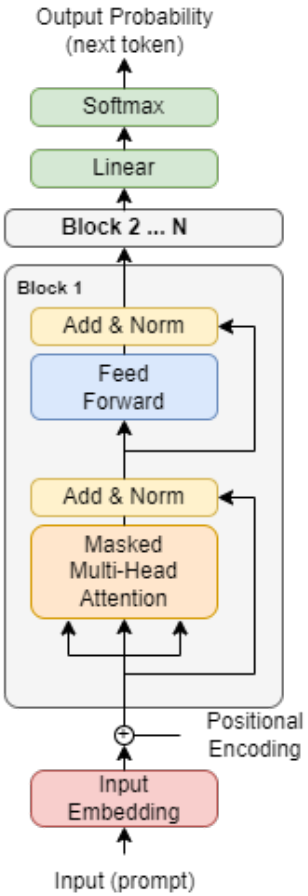


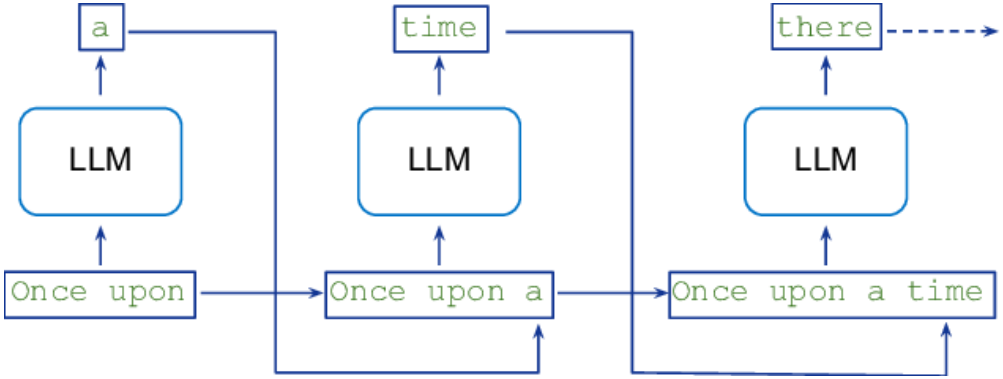




Large Language Models (LLMs)

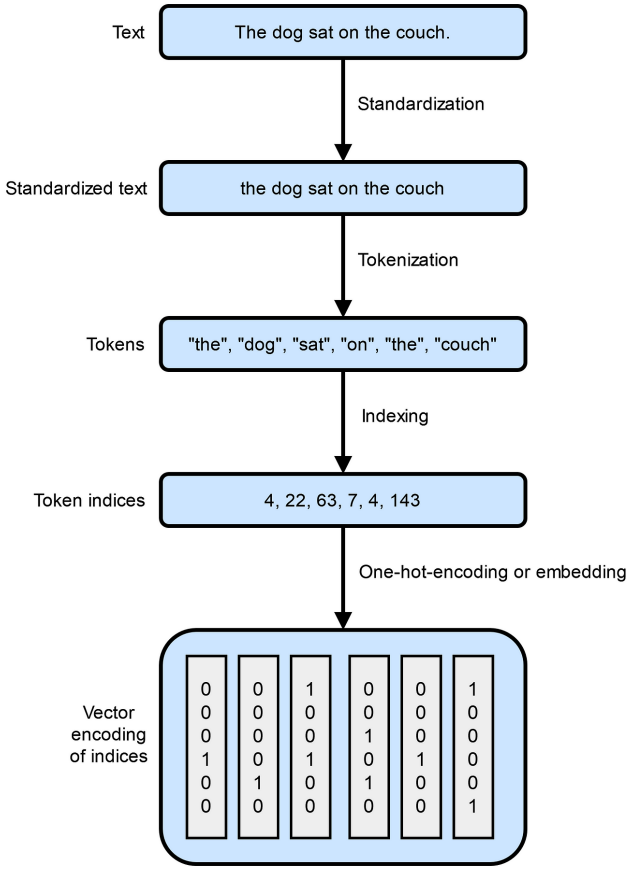
Large Language Models (LLMs) are **transformer**-based, **decoder-only** models trained using **autoregressive** next token prediction.





LLMs: Text tokenization

Mapping words to vectors.



LLMs: Numerical data tokenization

LLaMA 3 Tokenizer

- Digits: ['0', '1', ... '999']
- Token Ids: [15, 16, 17, ... 5500]

LLMTime

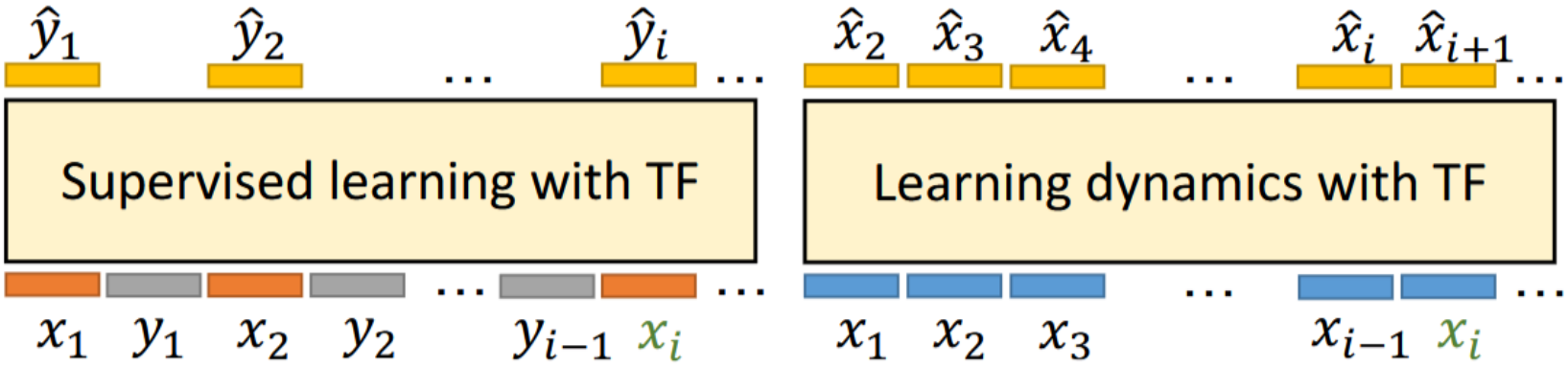
Time series processing

- Time series: [0.2513, 5.2387, 9.7889]
- Rescale+Encode: [150, 516, 850]
- Text: '150,516,850,'
- Tokens: ['150', ',', '516', ',', '850', ',']
- Token indices: [3965, 11, 20571, 11, 16217, 11]

Sampling: *Softmax* over the digits tokens' logits

In-context Learning (ICL)

In-context learning	Input prompt	Desired Output
Natural language processing	berry, baya, apple, manzana, banana	plátano
	Japan, mochi, France, croissant, Greece	baklava
Supervised learning $y_i = f(x_i) + \text{noise}$	$x_1, y_1, x_2, \dots, x_{i-1}, y_{i-1}, x_i$	$f(x_i)$
Dynamical systems $x_{i+1} = f(x_i) + \text{noise}$	$x_1, x_2, x_3, \dots, x_{i-2}, x_{i-1}, x_i$	$f(x_i)$



Transformers as Algorithms: Generalization and Stability in In-context Learning

Problem setup

Dynamics learning using LLMs

- State space $\mathbb{R}^{d_s}$, Action space $\mathbb{R}^{d_a}$, Reward $\mathbb{R}$
- Given a trajectory

$$\tau^{\pi} = (s_0, a_0, s_1, a_1, s_2, \ldots, s_{T-1})$$

We want to learn the distribution of the next state using ICL and a pre-trained LLM with parameters θ:

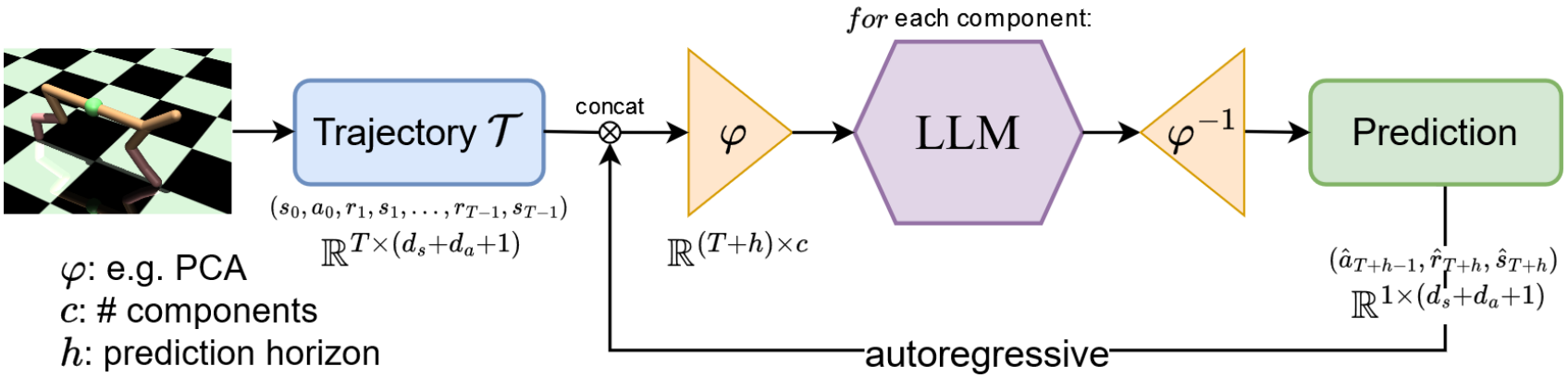
$$\{\hat{P}_{\theta}^{\pi,j}(s_t^j|\tau^{\pi})\}_{t\leq T, j\leq d_s} = \mathsf{ICL}_{\theta}(\tau^{\pi})$$

Challenges:

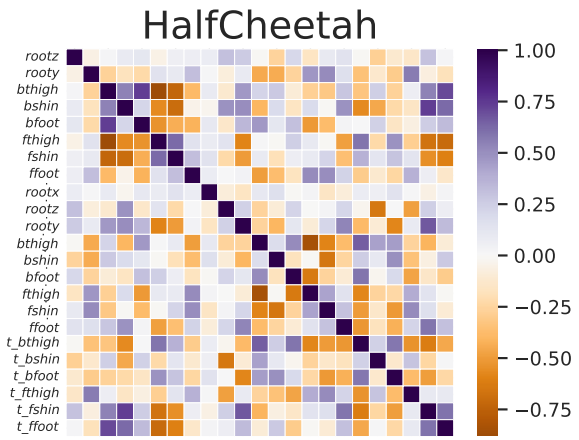
1. Multivariate states: $d_s > 1$
2. Including actions in-context: $P(s_t^j|s_0, a_0, s_1, a_1, s_2, \ldots, s_{T-1})$

Approach

DICL: Disentangled In-Context Learning [1]

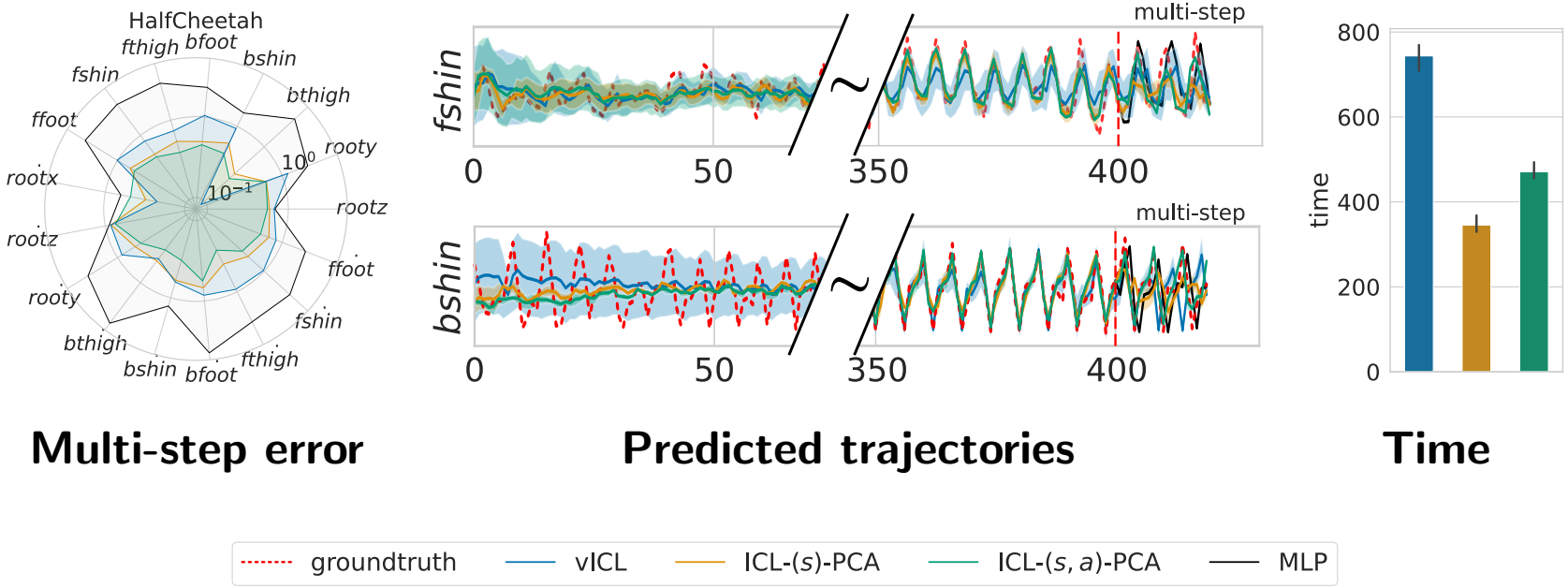


In practice, we project states and actions (s, a) into the space of PCA components.



Results

Results: Prediction Error



Multi-step error **Predicted trajectories** **Time**

PCA-based DICL achieves smaller multi-step error in less computational time. We compare DICL-(s) and DICL-(s, a) using a number of components equal to half the number of features, with the vanilla approach vICL and an MLP baseline.

Results: DICL-SAC

SAC: Soft Actor-Critic (an off-shelf RL algorithm)

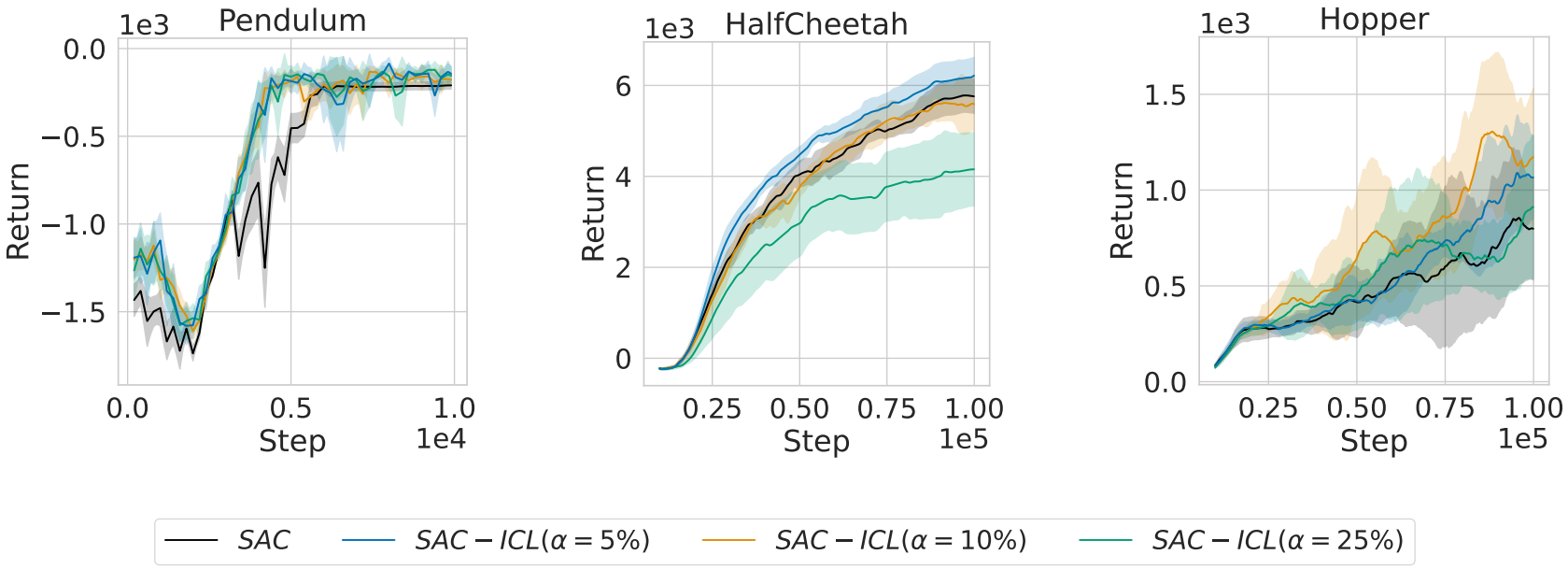
+

DICL

=

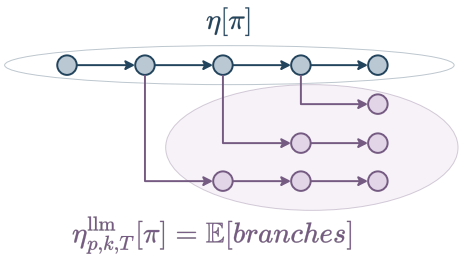
DICL-SAC

for $t = 1, \ldots, N_interactions$ **do**
  New transition (s_t, a_t, r_t, s_{t+1}) from π_θ
  Add (s_t, a_t, r_t, s_{t+1}) to $\mathcal{R}$
  Store auxiliary action $\tilde{a}_t \sim \pi_\theta(.|s_t)$
  if Generate LLM data **then**
    Sample trajectory $\mathcal{T} = (s_0, \ldots, s_{T_{\max}})$ from $\mathcal{R}$
    $\{\hat{s}_{i+1}\}_{0 \leq i \leq T_{\max}} \sim$ DICL-(s) $(\mathcal{T})$
    Add $\{(s_i, \tilde{a}_i, r_i, \hat{s}_{i+1})\}_{T \leq i \leq T_{\max}}$ to $\mathcal{R}_{\text{llm}}$
  end if
  if update SAC **then**
    Sample batch $\mathcal{B}$ of size b from $\mathcal{R}$
    Sample batch $\mathcal{B}_{\text{llm}}$ of size $\alpha \cdot b$ from $\mathcal{R}_{\text{llm}}$
    Update ϕ and ψ on $\mathcal{B} \cup \mathcal{B}_{\text{llm}}$
  end if
end for



Results: DICL-SAC (Theoretical guarantee)

Under mild assumptions on the LLM prediction error ε_{llm}, we have:



Theorem (Multi-branch return bound)

- T *the context length*
- $p \in [0, 1]$ *probability of branching*
- k *the branch length*
- ε_{llm} *the LLM in-context learning prediction error*

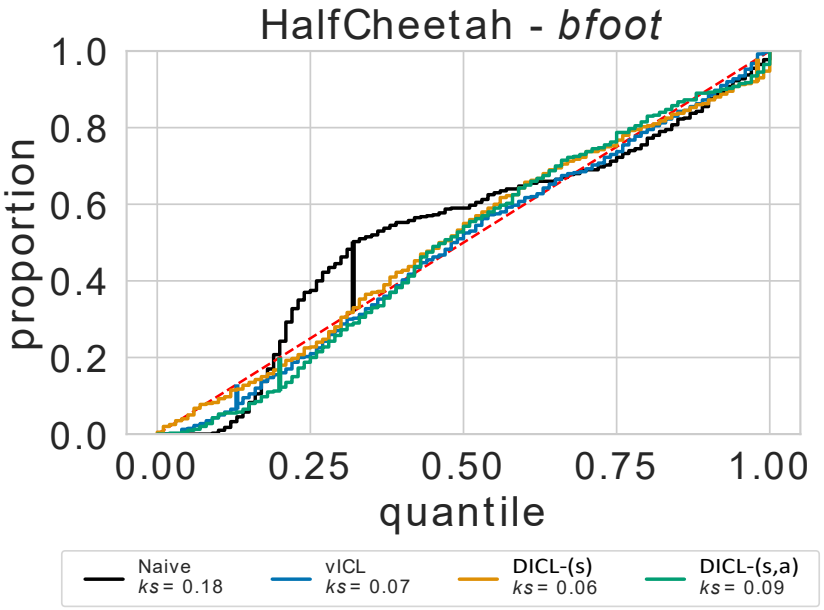
$$|\eta(\pi) - \eta^{llm}_{p,k,T}(\pi)| \leq 2\frac{\gamma^T}{1-\gamma} r_{\max} k^2 \, p \, \varepsilon_{llm}(T)$$

where $r_{\max} = \max_{s \in \mathcal{S}, a \in \mathcal{A}} r(s, a)$.

Results: Calibration

Quantile calibration: For probabilistic regression, a perfectly calibrated forecaster means that $p\%$ of groundtruth values fall within the $p\%$-confidence interval of the predicted CDF.

LLMs are well-calibrated in-context forecasters.



Question

How can we extend PCA to more complex learnable Adapters?

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Problem setup

Problem setup

Consider a multivariate time series forecasting task:

- $\mathbf{X} \in \mathbb{R}^{L \times D}$ data matrix
- $\mathbf{Y} \in \mathbb{R}^{H \times D}$ target
  - L lookback window (context length)
  - H forecasting horizon
  - D dimension (number of covariates)

We want to find the best adapter φ^* such that:

Definition (adapter)

Feature-space transformation $\varphi : \mathbb{R}^D \to \mathbb{R}^{D'}$ such that:

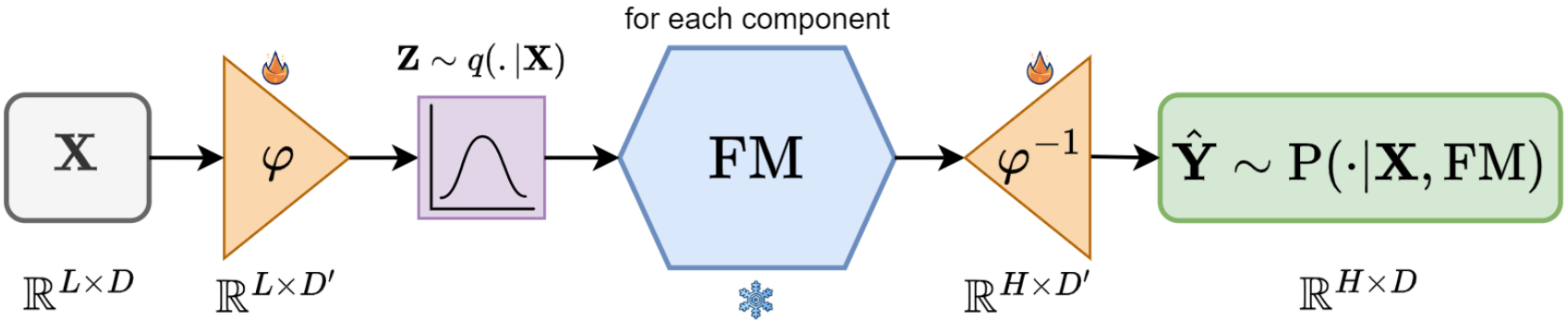
$$\hat{\mathbf{Y}}(\mathbf{X}; \varphi) = \varphi^{-1}\big(\texttt{FM}(\varphi(\mathbf{X}))\big), \text{ and } \varphi^* = \text{argmin}_{\varphi} \|\mathbf{Y} - \hat{\mathbf{Y}}(\mathbf{X}; \varphi)\|_{\mathrm{F}}^2,$$

where FM is a fixed time series foundation model.

Approach

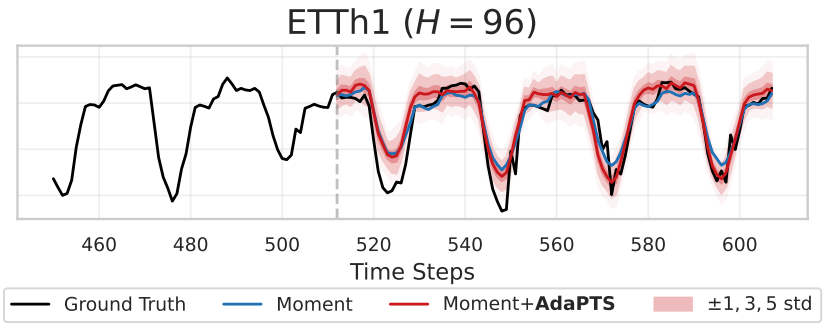
AdaPTS

AdaPTS: Adapters for Probabilistic multivariate Time Series forcasting [2]



Properties:

1. Mixing features
2. Probabilistic predictions



Theoretical analysis: Linear case

For a linear adapter $\varphi(\mathbf{X}) = \mathbf{X}\mathbf{W}_\varphi$ and a linear FM $f_{\mathrm{FM}}(\mathbf{X}) = \mathbf{W}_{\mathsf{FM}}^\top \mathbf{X} + \mathbf{b}_{\mathsf{FM}}\mathbf{1}^\top$, we have:

Proposition (Optimal linear adapter)

The closed-form solution of the problem

$$\min_{\mathbf{W}_\varphi} \mathcal{L}(\mathbf{W}_\varphi) = \|\mathbf{Y} - \left(\mathbf{W}_{FM}^\top \mathbf{X}\mathbf{W}_\varphi + \mathbf{b}_{FM}\mathbf{1}^\top\right)\mathbf{W}_\varphi^{-1}\|_F^2$$

writes as:

$$\mathbf{W}_\varphi^* = (\mathbf{B}^\top \mathbf{A})^+ \mathbf{B}^\top \mathbf{B},$$

where $\mathbf{A} = \mathbf{Y} - \mathbf{W}_{FM}^\top \mathbf{X}$ *and* $\mathbf{B} = \mathbf{b}_{FM}\mathbf{1}^\top$.

$\rightarrow$ Takeaway: The optimal solution is **not** the identity.

Results

Results: Forecasting error (MSE)

Families of adapters:

1. deterministic
   - Linear AutoEncoder
   - Deep non-linear AutoEncoder
2. probabilistic
   - + Variational Inference
   - + MC Dropout

Dataset	H	No adpt	with adapter				
		Moment	PCA	LinAE	dropLAE	LinVAE	VAE
ETTh1	96	$0.411_{\pm.012}$	$0.433_{\pm.001}$	$0.402_{\pm.002}$	$\mathbf{0.395_{\pm.003}}$	$0.400_{\pm.001}$	$0.404_{\pm.001}$
	192	$\mathbf{0.431_{\pm.001}}$	$0.440_{\pm.000}$	$0.452_{\pm.002}$	$0.446_{\pm.001}$	$0.448_{\pm.002}$	$\mathbf{0.431_{\pm.001}}$
Ill	24	$2.902_{\pm.023}$	$2.98_{\pm.001}$	$2.624_{\pm.035}$	$2.76_{\pm.061}$	$2.542_{\pm.036}$	$\mathbf{2.461_{\pm.008}}$
	60	$3.000_{\pm.004}$	$3.079_{\pm.000}$	$3.110_{\pm.127}$	$2.794_{\pm.015}$	$\mathbf{2.752_{\pm.040}}$	$2.960_{\pm.092}$
Wth	96	$0.177_{\pm.010}$	$0.176_{\pm.000}$	$0.169_{\pm.000}$	$\mathbf{0.156_{\pm.001}}$	$0.161_{\pm.001}$	$0.187_{\pm.001}$
	192	$0.202_{\pm.000}$	$0.208_{\pm.001}$	$\mathbf{0.198_{\pm.001}}$	$0.200_{\pm.001}$	$0.204_{\pm.000}$	$0.226_{\pm.000}$
ExR	96	$\mathbf{0.130_{\pm.011}}$	$0.147_{\pm.000}$	$0.167_{\pm.013}$	$\mathbf{0.130_{\pm.011}}$	$0.243_{\pm.039}$	$0.455_{\pm.010}$
	192	$\mathbf{0.210_{\pm.002}}$	$0.222_{\pm.000}$	$0.304_{\pm.005}$	$0.305_{\pm.013}$	$0.457_{\pm.020}$	$0.607_{\pm.021}$

Results: Interpretable latent representations

Desirable representation learning properties:

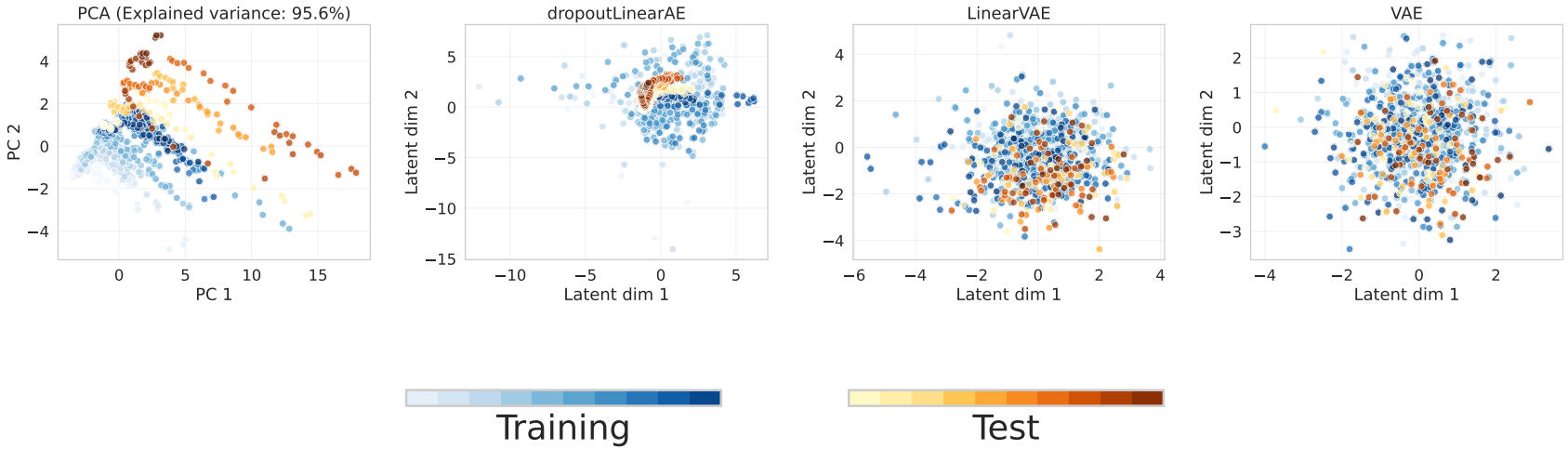
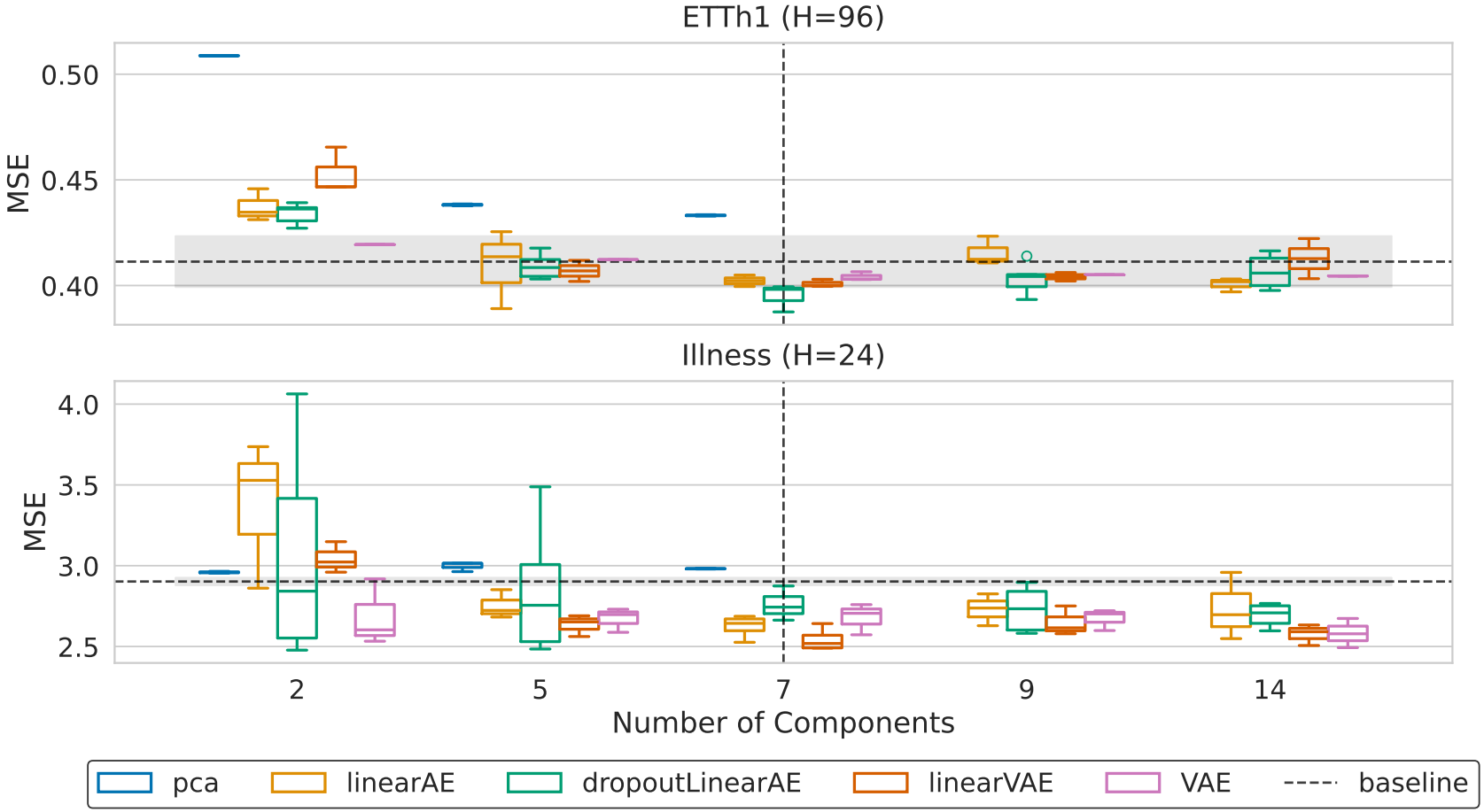


Figure: Visualization of the latent representation obtained by different adapters on Illness($H = 24$). Shaded colors indicate the time dimension, with lighter colors representing earlier timesteps.

Results: Dimensionality reduction

Better forecasting accuracy even with lower dimensions



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Conclusion

- We presented **DICL**, a methodology to adapt LLMs for the task of dynamics learning in MBRL
- We then presented **AdaPTS** a learning-based and probabilistic extension of adapters to multivariate time series forecasting

Take Home Message

Foundation Models are powerful predictors trained on vast amounts of data
$\rightarrow$ **Adapters** are an effective way to adapt FMs to custom problems

References

A. Benechehab, Y. A. E. Hili, A. Odonnat, O. Zekri, A. Thomas, G. Paolo, M. Filippone, I. Redko, and B. Kégl, "Zero-shot model-based reinforcement learning using large language models," in *The Thirteenth International Conference on Learning Representations (ICLR)*, 2025.

A. Benechehab, V. Feofanov, G. Paolo, A. Thomas, M. Filippone, and B. Kégl, "Adapts: Adapting univariate foundation models to probabilistic multivariate time series forecasting," *Forty-second International Conference on Machine Learning (ICML)*, May 2025.

Thank You!

Want to know more?

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Slides available at:

https://abenechehab.github.io/assets/pdf/research.pdf